

A $q > 2$ Deterministic-Dominator Square-Function Subcase

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Status

This packet records a narrow $q > 2$ partial result for Remark 12.10 of Yaroslavtsev's paper *Local characteristics and tangency of vector-valued martingales* [1]. The preceding packet

`1907.11588_Lq_p1_endpoint_qle2_square_function`

proves the $L^q(S)$, $p = 1$ endpoint for $1 < q \leq 2$. The full endpoint for $q > 2$ remains open here, but the square-function bottleneck is proved when the dominating square magnitudes are deterministic.

Source Problem

The source is arXiv:1907.11588, PDF page 105, Remark 12.10. The source states that the general local-martingale case is open and reduces the missing issue to an independent-increment comparison.

...and the fact that M and N are quasi-left continuous. $\square$

Remark 12.10. It is not known whether Theorem 12.8 holds for general local martingales. By Theorem 3.39 and by Proposition B.1 the main issue here is in proving a similar statement for discrete martingales with independent increments. Let us state this problem here as open. Let X be a Banach space, $1 \leq p < \infty$. Let $(\xi_n)_{n \geq 1}$ and $(\eta_n)_{n \geq 1}$ be X -valued mean-zero independent random variables such that for any Borel $B \in X \setminus \{0\}$

$$\sum_n \mathbb{P}(\xi_n \in B) \leq \sum_n \mathbb{P}(\eta_n \in B).$$

Does there exist a constant C (perhaps depending on p and X) such that

$$\mathbb{E} \left\| \sum_n \xi_n \right\|^p \leq C \mathbb{E} \left\| \sum_n \eta_n \right\|^p?$$

The continuation states the aggregate-domination formulation and notes that even the symmetric case was not known to the author.

By a standard symmetrization trick [66, Lemma 6.3] one can assume that ξ_n 's and η_n 's are symmetric. But even the symmetric case is not known for the author.

Main Square-Function Subcase

Let $q > 2$ and put $r = q/2 > 1$. The endpoint square-function comparison can be written at the positive $L^r(S)$ -level: for $X_i \in L^q(S)$, set $V_i = |X_i|^2 \in L^r(S)_+$. The desired quantity is

$$\mathbb{E} \left\| \sum_i V_i \right\|_{L^r(S)}^{1/2} = \mathbb{E} \left\| \left(\sum_i |X_i|^2 \right)^{1/2} \right\|_{L^q(S)}.$$

Theorem 1 (Deterministic square magnitudes). *Let $q > 2$, $r = q/2$, and let $v_1, \dots, v_m \in L^r(S)_+$ be deterministic. Let $V_1, \dots, V_n$ be independent $L^r(S)_+$ -valued random variables. Suppose that their aggregate occurrence measure is dominated by the deterministic square-magnitude measure:*

$$\sum_{i=1}^n \mathbb{P}(V_i \in A) \leq \sum_{j=1}^m \mathbf{1}_A(v_j), \quad A \in \mathcal{B}(L^r(S)), \quad 0 \notin A.$$

Then

$$\mathbb{E} \left\| \sum_{i=1}^n V_i \right\|_{L^r(S)}^{1/2} \leq C_q \left\| \sum_{j=1}^m v_j \right\|_{L^r(S)}^{1/2}.$$

Equivalently, if $u_j \in L^q(S)$ are deterministic and the aggregate laws of $|X_i|^2$ are dominated by the atoms $|u_j|^2$, then

$$\mathbb{E} \left\| \left(\sum_i |X_i|^2 \right)^{1/2} \right\|_{L^q(S)} \leq C_q \left\| \left(\sum_j |u_j|^2 \right)^{1/2} \right\|_{L^q(S)}.$$

Idea of the Proof

The $1 < q \leq 2$ packet used a positive multivariate Laplace representation. That representation fails for $q > 2$. In the deterministic-dominator subcase one can instead square first and use scalar Rosenthal pointwise in the spatial variable.

The scalar inequality controls

$$\mathbb{E} \left(\sum_i Z_i \right)^r$$

by the sum of a mean term and an r -moment term. Deterministic aggregate domination controls both terms pointwise:

$$\sum_i \mathbb{E} V_i \leq \sum_j v_j, \quad \sum_i \mathbb{E} V_i^r \leq \sum_j v_j^r \leq \left(\sum_j v_j \right)^r.$$

This is exactly the extra structure missing for a general random dominating family.

Proof

Lemma 1 (Scalar positive Rosenthal). *For every $r > 1$ there is a constant C_r such that for all finite independent nonnegative scalar random variables $Z_1, \dots, Z_n$,*

$$\mathbb{E} \left(\sum_i Z_i \right)^r \leq C_r \left[\left(\sum_i \mathbb{E} Z_i \right)^r + \sum_i \mathbb{E} Z_i^r \right].$$

Proof. For $1 < r \leq 2$, this is the von Bahr–Esseen inequality applied to $Z_i - \mathbb{E} Z_i$, followed by the triangle inequality in L^r . For $r > 2$, the usual scalar Rosenthal inequality gives

$$\mathbb{E} \left| \sum_i (Z_i - \mathbb{E} Z_i) \right|^r \leq C_r \left[\left(\sum_i \mathbb{E} Z_i^2 \right)^{r/2} + \sum_i \mathbb{E} Z_i^r \right].$$

Let $A = \sum_i \mathbb{E} Z_i$ and $B = \sum_i \mathbb{E} Z_i^r$. Interpolation and Holder's inequality give

$$\sum_i \mathbb{E} Z_i^2 \leq A^{(r-2)/(r-1)} B^{1/(r-1)}.$$

Hence

$$\left(\sum_i \mathbb{E} Z_i^2 \right)^{r/2} \leq A^r + B$$

by Young's inequality. Combining this with $\|\sum_i Z_i\|_{L^r} \leq A + \|\sum_i (Z_i - \mathbb{E} Z_i)\|_{L^r}$ proves the lemma. $\square$

Proof of Theorem 1. Let

$$A = \sum_{j=1}^m v_j \in L^r(S)_+.$$

The aggregate domination hypothesis implies the following two lattice inequalities:

$$\sum_i \mathbb{E} V_i \leq A \quad \text{in } L^r(S), \quad \sum_i \mathbb{E} V_i^r \leq \sum_j v_j^r \quad \text{in } L^1(S).$$

Indeed, both follow by testing against nonnegative simple functions and using the aggregate domination for the Borel maps

$$V \mapsto \int_S h V \, d\sigma, \quad V \mapsto \int_S h V^r \, d\sigma$$

with $h \geq 0$.

For almost every $s \in S$, apply Lemma 1 to the independent nonnegative scalar variables $V_i(s)$. Then

$$\mathbb{E} \left(\sum_i V_i(s) \right)^r \leq C_r \left[\left(\sum_i \mathbb{E} V_i(s) \right)^r + \sum_i \mathbb{E} V_i(s)^r \right].$$

Integrating over S gives

$$\begin{aligned} \mathbb{E} \left\| \sum_i V_i \right\|_{L^r}^r &\leq C_r \left[\int_S \left(\sum_i \mathbb{E} V_i \right)^r \, d\sigma + \int_S \sum_i \mathbb{E} V_i^r \, d\sigma \right] \\ &\leq C_r \left[\int_S A^r \, d\sigma + \int_S \sum_j v_j^r \, d\sigma \right] \leq 2C_r \|A\|_{L^r}^r, \end{aligned}$$

because $\sum_j v_j^r \leq (\sum_j v_j)^r = A^r$.

Finally, by Jensen's inequality for the concave map $t \mapsto t^{1/(2r)}$,

$$\mathbb{E} \left\| \sum_i V_i \right\|_{L^r}^{1/2} = \mathbb{E} \left(\left\| \sum_i V_i \right\|_{L^r}^r \right)^{1/(2r)} \leq \left(\mathbb{E} \left\| \sum_i V_i \right\|_{L^r}^r \right)^{1/(2r)} \leq (2C_r)^{1/(2r)} \|A\|_{L^r}^{1/2}.$$

This is the asserted estimate. $\square$

Discrete Martingale Corollary

Proposition 1 (Random signs with deterministic magnitudes). *Let $q > 2$. Let $u_1, \dots, u_m \in L^q(S)$ be deterministic and let $\eta_j = \varepsilon_j u_j$, where (ε_j) are independent Rademacher signs. Let (ξ_i) be independent mean-zero $L^q(S)$ -valued random variables. Assume that the aggregate square-magnitude laws satisfy*

$$\sum_i \mathbb{P}(|\xi_i|^2 \in A) \leq \sum_j \mathbf{1}_A(|u_j|^2), \quad A \in \mathcal{B}(L^{q/2}(S)), \quad 0 \notin A.$$

Then

$$\mathbb{E} \max_{1 \leq k \leq n} \left\| \sum_{i=1}^k \xi_i \right\|_{L^q(S)} \leq C_q \mathbb{E} \max_{1 \leq k \leq m} \left\| \sum_{j=1}^k \eta_j \right\|_{L^q(S)}.$$

Proof. Theorem 1 gives

$$\mathbb{E} \left\| \left(\sum_i |\xi_i|^2 \right)^{1/2} \right\|_{L^q} \leq C_q \left\| \left(\sum_j |u_j|^2 \right)^{1/2} \right\|_{L^q}.$$

The right-hand side is, deterministically,

$$\mathbb{E} \left\| \left(\sum_j |\eta_j|^2 \right)^{1/2} \right\|_{L^q}.$$

Yaroslavtsev's Banach-function-space BDG theorem [2, Theorem 4.32 and Proposition 4.33] compares the L^1 maximal norm of each martingale difference sequence with its lattice square function. Applying it to (ξ_i) and (η_j) proves the claim. $\square$

What This Does Not Yet Prove

The full $q > 2$ endpoint would require the same square-function comparison when the dominating family has random square magnitudes $W_j = |Y_j|^2$. The argument above gives, for a dominated family V_i ,

$$\mathbb{E} \left\| \sum_i V_i \right\|_{L^r}^{1/2} \leq C_r \left[\int_S \left(\sum_j \mathbb{E} W_j \right)^r d\sigma + \int_S \sum_j \mathbb{E} W_j^r d\sigma \right]^{1/(2r)}.$$

For deterministic $W_j = v_j$, this is controlled by $\left\|\sum_j v_j\right\|_{L^r}^{1/2}$. For random W_j , however, the target is

$$\mathbb{E} \left\| \sum_j W_j \right\|_{L^r}^{1/2},$$

and rare large jumps show that this small moment need not control the displayed r -moment Rosenthal functional. Thus this packet isolates a specific anti-concentration or de-Poissonization step as the remaining obstruction for this route.

Stress Tests and Failed Tickets

During this push I also tested finite-dimensional regrouping patterns in ℓ_q^d , including the c_0 -style occupancy construction with fixed finite q . The simple occupancy ratios stayed bounded, as expected: for deterministic disjoint atoms, multinomial counts have uniformly bounded r -moments for each fixed $r = q/2$. Random sparse/correlated atom searches for $r = 1.5$ and $r = 2$ found only ratios around 1.06 and 1.13, not dimension growth. These computations are not proof, but they support the view that the $q > 2$ obstruction is an anti-concentration proof issue rather than the elementary c_0 occupancy obstruction.

Two proof tickets remain plausible:

1. a positive-vector de-Poissonization theorem comparing $\mathbb{E} \|\sum_i V_i\|_r^{1/2}$ with the compound-Poisson model determined only by the aggregate measure;
2. a lower bound showing that every independent dominating family controls the correct truncated Rosenthal functional, not the too-large r -moment functional displayed above.

Novelty Notes and Human Review

The run indexes contain the $1 < p \leq L^q$ packet, the $1 < q \leq 2$ endpoint packet, Hilbert packets, the c_0 counterexample, and the conditional endpoint reduction. No packet in the run records this deterministic dominating-magnitude $q > 2$ subcase.

Human review should first check the lattice consequences of aggregate domination:

$$\sum_i \mathbb{E} V_i \leq \sum_j v_j, \quad \sum_i \mathbb{E} V_i^r \leq \sum_j v_j^r.$$

After that, the rest is the scalar Rosenthal inequality and Jensen's inequality.

References

- [1] I. S. Yaroslavl'tsev. Local characteristics and tangency of vector-valued martingales. arXiv:1907.11588, 2019.
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- [4] H. P. Rosenthal. On the subspaces of L_p , $p > 2$, spanned by sequences of independent random variables. *Israel Journal of Mathematics* 8 (1970), 273–303.
- [5] S. J. Montgomery-Smith and A. R. Pruss. A comparison inequality for sums of independent random variables. *arXiv:math/9811124*, 1998.